

* Evolution *

Term Evolution was given by Herbert Spencer

Origin of Universe

① The Big Bang Theory - Given by Abbe Lemaitre

→ Huge single explosion



Gases dissipated



Gravitation pull

→ Hot → celestial body (Sun)

→ Condense → Planets

→ Hydrogen and Helium formed sometime later

→ In solar system of Milky Way galaxy, earth was supposed to have been formed about 4.5 billion years ago.

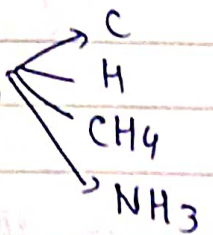
→ Origin of universe - 20 billion years ago

→ Evolution of Earth

4.5 billion years ago → primitive earth

→ High Temperature

→ No atmosphere



→ Life originated 4 billion years ago

* Origin of life

Theories of origin of life

① Panspermia theory or Cosmic Theory

→ Early Greek thinkers thought units of life called spores were transferred to different planets including earth

→ Panspermia is still a favourite idea for some astronomers

② Theory of Special creation

→ Earth is 4000 years old

→ Organisms were present as it is

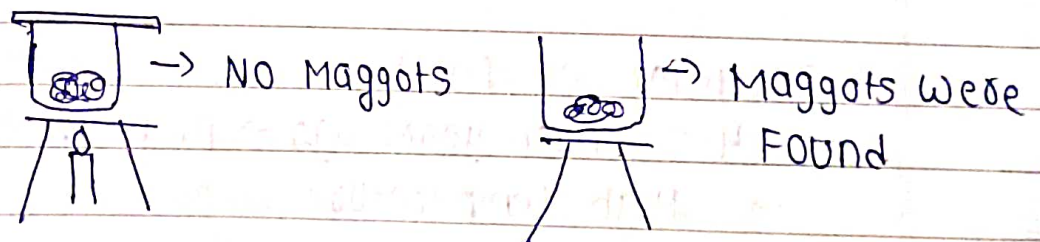
③ Theory of Spontaneous Generation

→ Given by von Helmont

→ Stated that life came out of decaying and rotting matter like, straw, mud etc

→ Disapproved

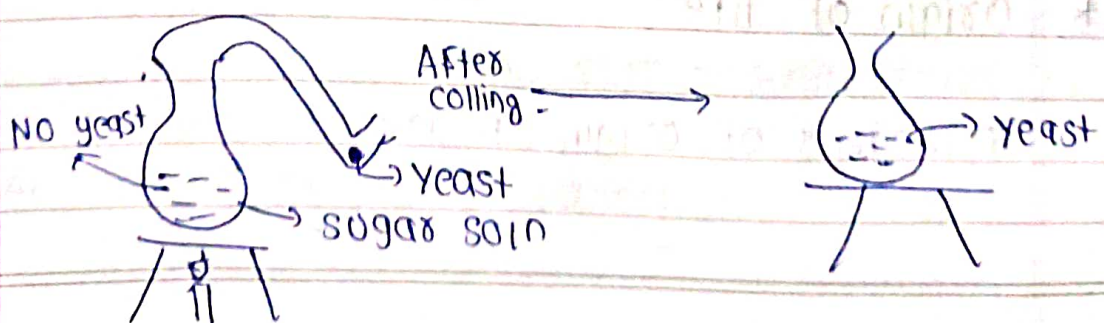
Reddi's experiment



Disapproved theory of Spontaneous Generation

④ Theory of Biogenesis → Huxley

→ Louis pasteur



Oparin Haldane Hypothesis

- Chemical Evolution
- Biological Evolution

Primitive earth → Unstable
— High Temp
— Lightening
- Atmosphere is Reducing
- Aquatic Habitat

→ Gases are - C, H, N, O → It laid to formation of CH_4 , NH_3 + HCN
Molecular stage

→ Formation of Simple Micromolecules
Glucose, Amino acid, N Base, Pentose, Sugar, Phosphate group

→ Formation of Complex Micromolecules

Carbohydrates	}	Goes into	Water (Hot dilute Soup)
Proteins			
Lipids			
Nucleic acid			

→ Some of these micromolecules + macromolecules combined and Gained the ability to grow and divide

- | | | |
|-----------------------------|---|------------|
| ① Oparin → Coacervates | } | A cellular |
| ② Sidney Fox → Microspheres | | |

→ Formation of prokaryotes

→ Formation of prokaryotes

Heterotrophs

Chemoautotrophs

Photoautotrophs



O₂ → The earth atmosphere became oxidizing

Formn of ozone layer → Atmosphere stable

→ Eukaryotes

First non-cellular life forms could be have originated 3 billion years ago and they were RNA, protein, and polysaccharide

* Evolution of life forms

Charles Darwin

→ Theory of Natural Selection

* Island of Galepagos

* Darwin Finches

* Branching descent and natural selection

Theory of Natural Selection

i) → survival of the fittest

Competition → [Intra specific competition
↳ Inter specific competition

ii) Natural Selection

iii) Origin of species

iv) slow & Gradual variation

→ Similar conclusion of Darwin ~~of~~ was seen by Alfred Wallace

↳ Worked on Malay Archipelago (Now in Indonesia)

* Evidences for Evolution.

Palaeontological Evidences

→ Study of Fossil

→ Tracing evolutionary History.

→ Connecting link

→ Study extinct animals

→ Determine geological period

Missing link

↳ Archaeopteryx → missing link betn Reptiles & Birds

- Both characters are present

- Bird like Reptiles

① Feathers ① Skin scaly

② Beak ② Tail

③ wing ③ Teeth

④ Non pneumatic bones

Embryological Evidences.

→ Ernst Haeckel

→ Common features during development

→ Karl Ernst von ~~Baer~~ Baer

→ Ontogeny repeats phylogeny - Embryonic structure mimics the course of Evolution

Biochemical Evidences

→ Similarities of proteins, genes suggest common ancestry in organisms

Morphological and Anatomical Evidences

Homologous Organs

→ Similarity in structure and different in function
→ Common origin and ancestor. Divergent evolution
→ Related species become less similar

→ Examples of Homologous organs

① Forelimbs of mammals - Whale, Bat, Cheetah, Human

② Thorns of Bougainvillea and tendrils of cucurbits
(Both are modification of axillary bud)

③ Vertebrae Heart & Brain

④ Mouth parts of insect

⑤ Potato and Ginger

Analogous Organs

→ Similarity in function but different in structure.

→ Common Habitat, point of origin is different.

→ Common function. Convergent evolution

→ Unrelated species become more similar

Example of analogous organs

① wings of Butterfly and Bird.

② Eye of octopus and mammal.

③ Flipper of penguin & dolphin

④ Sweet potato and potato

⑤ Sting of Bee and Scorpion.

* Adaptive Radiation

→ A kind of divergent evolution.

The process of evolution of different species in a given geographical area starting from a point and literally radiating other areas.

eg → Darwin Finches & Australian Marsupials

Darwin Finches

Original - seed eating variety.

beak size / structure → changed

Australian Marsupials (Pouched Animal)

When more than one Adaptive radiation occurs in a geographical area; the two of them are said to be having convergent evolution.

Evidences from Vestigial Organ.

→ Nictitating membrane

→ muscles of pinna

→ Vermiform appendix

→ coccyx

→ Third molar

→ Body Hair

→ segmented muscles of abdomen.

* ATAVISM - Sudden reappearance of certain extinct features in organism. Ex - ① Tail in born baby

② pointed canine teeth, ③ Extra nipple.

④ Excessive body hair.

Evidences from connecting link:

- Between Non living & living → Virus
- Plant & Animal → Euglena
- Annelida & Mollusca → Neopalina
- Annelida & Arthropoda → Pexipatis
- Non chordata & chordata → Balanoglossus
- Bony & cartilaginous Fish → Chimera

* Lamarckism:

→ According to Lamarck, evolution of life forms had occurred and was driven by the use and disuse of organs

- ① Every organism has internal vital force
- ② Effect of environment and new needs
- ③ Use and disuse of organ
- ④ Inheritance of Acquired traits

→ August Weismann - Theory of continuity of Germ plasma.

* Mutational Theory

Hugo de Vries - work on evening primrose
(*Oenothera lamarckiana*)

Mutation - sudden

- Random

- direction less

- large

→ Single Step large mutation → Saltation

HUGO de Vries	Darwin
→ The mutation causes evolution.	→ Minor heritable variation causes evolution.
→ Mutation are random & directionless.	→ Variation are small & directional.
→ The mutation causes speciation (saltation).	→ Evolution for Darwin was gradual.

* Evolution Types

- ① Progressive → Simple to Complex.
- ② Retrogressive → Complex to Simple.

* A Brief account of Evolution

1) Azoic era	Age in million years from present 4600	The era of no life.
2) Archaeozoic era	3500 million years	The era of invisible life.
3) Protozoic era	3000-1000 million years	The era of early life.
4) Palaeozoic era	570-280 million yrs	The era of ancient life.
i) Cambrian		Age of trilobites.
ii) Ordovician		Age of invertebrates.
iii) Silurian		Origin of jawed fishes.
iv) Devonian		Age of Fishes.
v) Carboniferous		Age of amphibians.
vi) Permian		Trilobites become extinct.

5)	Mesozoic era	225-135 million years	The era of medieval life Origin of dinosaur Dinosaur become abundant Dinosaur become extinct
	i) Triassic		
	ii) Jurassic		
	iii) Cretaceous		
6)	Cenozoic era	65-0.01 million years	Divided into tertiary and quaternary periods
7)	Tertiary period	65-26 million years	Rise of first primates Diversification happened Rise of first monkeys Mammals were at peak Age of mammals
	i) Paleocene epoch		
	ii) Eocene epoch		
	iii) Oligocene epoch		
	iv) Miocene epoch		
	vi) Pliocene epoch		
8)	Quaternary Period		
	i) Pleistocene epoch	2 million years	Great mammals extinct
	ii) Holocene	0.01 million years	Modern man is dominant

# Events	Time Scale
1) First cellular form of life on earth	2000 million years ago
2) Invertebrates were formed & active	500 mya
3) Jawless fish evolved	350 mya
4) Seaweeds and a few plants existed	320 mya
5) Fish with stout & strong fins	350 mya
6) Fish like reptiles (eg- Ichthyosaurs)	200 mya
7) Dinosaurs disappeared from earth	65 mya

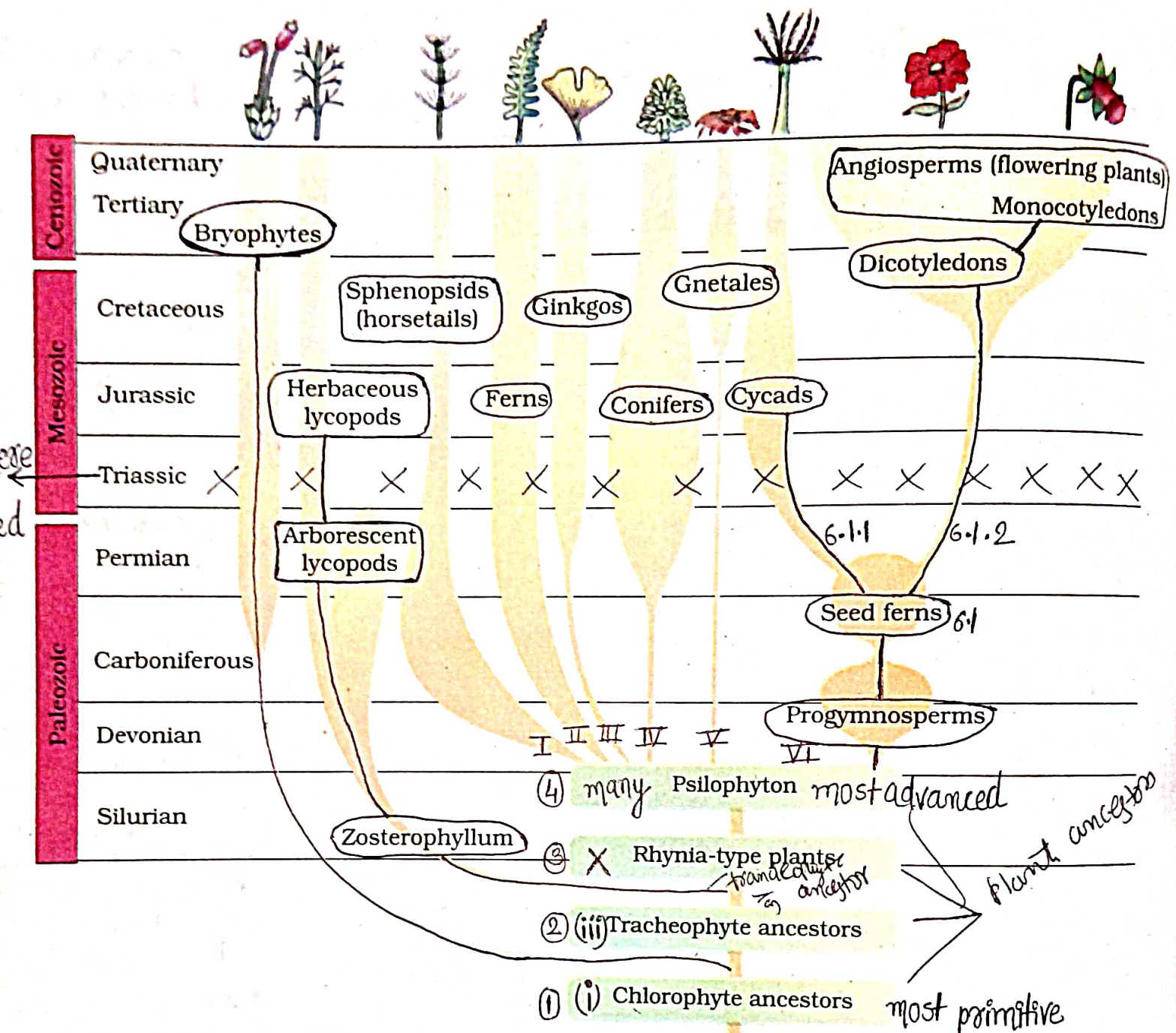


Figure 7.9 A sketch of the evolution of plant forms through geological periods

EVOLUTION

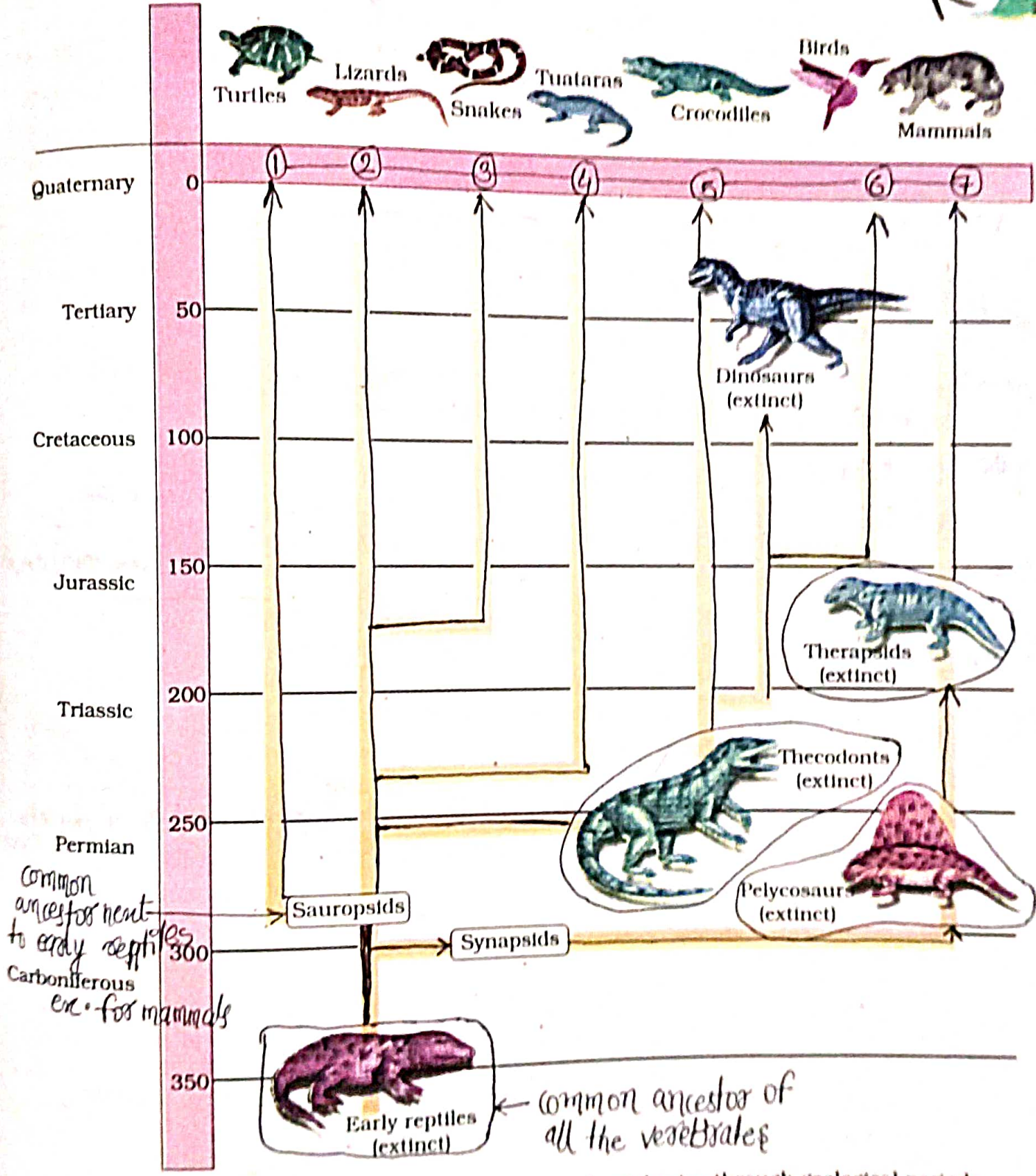


Figure 7.10 Representative evolutionary history of vertebrates through geological periods

* Origin and Evolution of Man

^{Ape like} Oxyopithecus & ^{Man like} Ramapithecus → Man like primates →
 Australopithecines → Homo Habilis → Homo erectus ↓

Homo sapiens ← Homo sapiens fossil ← Neanderthal man

Genus	Age of appearance	Characteristic Feature	Brain capacities
Oxyopithecus & Ramapithecus	15 mya	→ Hairy and walk like Gorillas & chimpanzees	
Man like primates	3-4 mya	→ Not taller than 4 feet but walked upright → Fossils of man-like bones have been discovered in Ethiopia & Tanzania	
Australopithecines (Taung child)	2 mya	→ Lived in East African Grassland → Hunted with stone weapons & ate Fruit	500 cc
Homo Habilis (first human like being)		→ The toolmaker handyman → Did not eat meat	650-800 cc
Homo erectus (the erect man)	1.5 mya	→ Fossils discovered in Java in 1891 → Ate meat	900 cc

Neanderthal man	1,00,000 - 40,000 years back	→ Lived in near east and central Asia. → Used hides to protect their body and buried their dead	1400 cc.
Homo sapiens Fossils	34000 years ago	Arose in Africa	1650 cc
Homo sapiens Sapiens	75,000 - 10,000 years ago (ice age)	→ Backbone with 4 curves cultivate the plant and domesticate animals.	1450 cc

→ Pre-historic cave art developed 18,000 years ago. Evidence is available at Bhimbetka Rock Shelter in Raisen district of Madhya Pradesh.

→ Agriculture and human settlements started 10,000 years back.

* Hardy - Weinberg Principle

Sum of Allelic frequency in a gene pool is constant

Factors that affect Hardy - Weinberg's Equilibrium

- ① Gene migration or gene flow
- ② NO Genetic Drift (sudden change in allelic frequency)
- ③ NO Mutation
- ④ NO Genetic recombination
- ⑤ NO Natural selection.

P → dominant Allele

Q → Recessive allele frequency

$$(p + q) = 1$$

$$(p + q)^2 = 1^2$$

$$p^2 + 2pq + q^2 = 1$$

AA → Homozygous dominant Genotype (p^2)

Aa → Heterozygous " " ($2pq$)

aa → Homozygous recessive Genotype (q^2)